

Nicholai Martinsen

San Francisco Bay Area, California

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SUMMARY

Full-stack software engineer with production experience architecting, building, and deploying modern web applications, internal tools, and containerized infrastructure from the ground up. Skilled across backend systems, database architecture, React/TypeScript frontends, Dockerized deployments, and CI/CD. Self-directed AI systems builder with hands-on experience running local LLMs, embedding/reranking services, agent frameworks, speech-to-text/text-to-speech models, and memory/retrieval systems using PostgreSQL, vector search, and graph-based approaches.

WORK EXPERIENCE

Software Architect – Perseeva | June 2025 – Present

Own frontend modernization, Docker infrastructure, deployment workflows, and production POC reliability for an IoT/BMS platform processing live BACnet device data across Cisco switch infrastructure.

- Took over critical architecture responsibilities across deployment infrastructure, frontend modernization, and production readiness.
- Replaced uncontainerized deployment process involving bash scripts, systemd services, and manual RocksDB compilation with Docker-based infrastructure for repeatable customer deployments.
- Pre-compiled RocksDB into containerized database agents, reducing deployment friction from hours to minutes and eliminating client-side compilation requirements.
- Supported first production POC at a major international airport, processing live BACnet device data across Cisco switch infrastructure.
- Built management console frontend in React 19, TypeScript, TanStack Router, Vite 7, and Tailwind CSS v4 with JWT authentication and protected routes.
- Designed Docker-in-Docker development environment with 5+ services and GitHub Actions CI/CD for local development, testing, and automated image builds.
- Integrated Sparkplug B messaging patterns to standardize IoT data flow between edge agents, backend services, and management tooling.

Software Developer (Full-stack) – Contra Costa County Public Works | July 2024 - April 2026

Architecting and delivering internal tools for county-wide capital project management

- Designed and deployed a production internal tooling platform that replaced manual Word/email-based workflows with a secure database-driven application for capital project and real-estate request management across 20 county departments and 200+ users.
- Replaced manual Word-based submission process with a secure, database-driven workflow, eliminating print-and-sort overhead for hundreds of annual submissions
- Led backend migration from Node.js to Go to improve maintainability, permissions modeling, and scalability for multi-division internal workflows.
- Designed SQL Server database with stored procedures, triggers, and historical auditing
- Built Dockerized development and production environments with GitHub Actions CI/CD
- Created custom SQL migration tool for versioned schema updates across environments
- Implemented Redis caching and optimized backend performance for multi-department usage

Self-Employed Contractor | June 2020 – October 2025

- Managed client relationships, project planning, and execution for residential remodeling projects
- Developed strong problem-solving, estimation, and client management skills

SELECTED TECHNICAL PROJECTS

Local AI Systems & Agent Memory Lab

Self-directed experimentation with local LLM infrastructure, retrieval systems, and agent frameworks.

- Built and maintained a local Docker-based AI stack with services for LLM inference, embeddings, reranking, speech-to-text, and text-to-speech experimentation.
- Ran and compared local CPU/GPU inference workflows, including smaller mixture-of-experts models, quantized models, and OpenAI-compatible local endpoints.
- Integrated Parakeet v3 speech-to-text into local desktop shortcuts for audio transcription and direct text entry workflows.
- Experimented with long-term AI memory systems using PostgreSQL-backed storage, vector retrieval, reranking, and graph-based memory approaches.
- Evaluated agent frameworks including OpenClaw, AgentZero, Letta, and Hindsight for tool use, persistent memory, retrieval quality, and local-first AI workflows.
- Modified and configured open-source AI tools to test retrieval behavior, memory persistence, context injection, and agent orchestration patterns.

EDUCATION

Bachelor's Degree in Computer Science | California State University Monterey Bay (Fall 2025)

Associate's Degree in Computer Science | Diablo Valley College (Spring 2023)

Certifications: IBM z/OS Mainframe Practitioner • Enterprise Design Thinking Practitioner • Agile Explorer • IBM z Xplore • New Collar Professional Skills

TECHNICAL SKILLS

Languages & Frameworks: Go, TypeScript, React 19, Node.js, Express, Python, SQL

Databases & Storage: SQL Server, PostgreSQL, MySQL, MongoDB, Redis, vector search concepts

AI / LLM Systems: Local LLM inference, embeddings, reranking, RAG, prompt workflows, agent frameworks, speech-to-text, text-to-speech

Infrastructure: Docker, Docker-in-Docker, GitHub Actions CI/CD, Linux, Nginx, OpenAI-compatible local services

Architecture: REST API design, JWT authentication, microservices, internal tools, IoT data pipelines, AI memory/retrieval systems
